

Exam in Numerical Analysis E3, FMN050, 060602

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This exam lasts 08:00–11:00. In order to pass, a minimum of 15 points is required on the exam. To this score your computer assignment score is added. The max total score on computer assignments + exam is 50 points. The final grade of the course will be: for **grade 3**, 25–33 points are required; **grade 4**: 34–42 p, **grade 5**: 43–50 p (preliminary figures).

During the exam you are allowed to use a pocket calculator. No other material of any kind (e.g. textbook, lecture notes, electronic material, etc.) is allowed. Please feel free to answer the questions in Swedish or in English.

Problems

A. Condition and Error analysis

1. According to Newton, the force F between two masses, m_1 and m_2 , at a distance r is

$$F = G \cdot \frac{m_1 m_2}{r^2}.$$

Determine the relative uncertainty in F when we know that $G = 6.67 \cdot 10^{-11}$, while m_1 and m_2 are known to a relative accuracy of 10^{-4} , and r is known to 5 digits. What is the source of the biggest error? **(1p)**

Solution. Logarithmic differentiation yields

$$\frac{dF}{F} = \frac{dG}{G} + \frac{dm_1}{m_1} + \frac{dm_2}{m_2} - 2\frac{dr}{r}$$

from which it follows that

$$\frac{\Delta F}{F} \leq \frac{0.005}{6.67} + 2 \cdot 10^{-4} + 2 \cdot 5 \cdot 10^{-5}.$$

(Note that “known to 5 digits” means that the relative error is, in the worst case, $5 \cdot 10^{-5}$, as is seen from the example $r = 1.0001$ which is known to 5 digits if the absolute error is at most 0.00005.) Hence

$$\frac{\Delta F}{F} \leq 8 \cdot 10^{-4} + 2 \cdot 10^{-4} + 10^{-4} = 1.1 \cdot 10^{-3}$$

and the biggest error contribution obviously comes from not knowing the constant G to better accuracy. (This contributes some 70% of the error.)

2. Let

$$A = \begin{bmatrix} 3 & 1 & 1 \\ -1 & 0 & 1 \\ 4 & 2 & 0 \end{bmatrix}; \quad A^{-1} = \frac{1}{4} \begin{bmatrix} 2 & -2 & -1 \\ -4 & 4 & 4 \\ 2 & 2 & -1 \end{bmatrix}$$

Compute the condition number of this matrix with respect to the max norm. How much larger can the solution's relative error $\|\delta x\|_\infty/\|x\|_\infty$ be compared to the data error $\|\delta b\|_\infty/\|b\|_\infty$ when solving $Ax = b$? **(1p)**

Solution. $\|A\|_\infty = 6$ and $\|A^{-1}\|_\infty = 12/4$, so $\kappa_\infty[A] = 6 \cdot 3 = 18$. Therefore the relative error can be at most 18 times bigger than the data error.

3. Suppose that in the system above $b = (0 \quad 2.2305 \quad 0)^T$, where the zero elements are exact and contain no error, while the middle component is correctly rounded. How much larger is the solution's *maximum relative error*, measured in the max norm in this case? **(1p)**

Solution. If the first and last components of b are always zero, we have

$$x = \frac{1}{4} \begin{bmatrix} 2 & -2 & -1 \\ -4 & 4 & 4 \\ 2 & 2 & -1 \end{bmatrix} \begin{bmatrix} 0 \\ b_2 \\ 0 \end{bmatrix} = b_2 \begin{bmatrix} -1/2 \\ 1 \\ 1/2 \end{bmatrix}$$

So $\|x\|_\infty = |b_2|$ and $\|\delta x\|_\infty = |\delta b_2|$; therefore the max relative error (in the max norm) is at most as big as the relative error in b_2 .

B. Interpolation and Least squares

1. According to the US Department of Energy, the total market for Jet A-1 fuel in the Far East and Oceania has developed according to Table 1 (unit is million gallons per day).

1990	1991	1992	1993	1994	1995	1996	1997	1998
20.2	21.1	25.5	28.9	30.7	36.0	36.1	38.7	35.2

Table 1: Data for problem **B1**.

a) Fit a straight line to the data. **(2p)**

b) Fit an exponential function to the same data. **(2p)**

Solution. Center the straight line around $t = 1994$ and write the line $P(t) = c_0 + c_1(t - 1994)$. We get the system

$$\begin{bmatrix} 1 & -4 \\ 1 & -3 \\ 1 & -2 \\ 1 & -1 \\ 1 & 0 \\ 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \end{bmatrix} \begin{bmatrix} c_0 \\ c_1 \end{bmatrix} = \begin{bmatrix} 20.2 \\ 21.1 \\ 25.5 \\ 28.9 \\ 30.7 \\ 36.0 \\ 36.1 \\ 38.7 \\ 35.2 \end{bmatrix}$$

from which we derive the normal equations

$$\begin{bmatrix} 9 & 0 \\ 0 & 60 \end{bmatrix} \begin{bmatrix} c_0 \\ c_1 \end{bmatrix} = \begin{bmatrix} 272.4 \\ 141.1 \end{bmatrix}$$

with solution $c_0 = 30.3$ and $c_1 = 2.35$. Fitting an exponential function follows the same pattern. Put $y = a_0 e^{a_1(t-1994)}$ and fit $\log y = \log a_0 + a_1(t - 1994)$ to the logarithm of the data. We get

$$\begin{bmatrix} 1 & -4 \\ 1 & -3 \\ 1 & -2 \\ 1 & -1 \\ 1 & 0 \\ 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \end{bmatrix} \begin{bmatrix} \log a_0 \\ a_1 \end{bmatrix} = \begin{bmatrix} 3.0057 \\ 3.0493 \\ 3.2387 \\ 3.3638 \\ 3.4243 \\ 3.5835 \\ 3.5863 \\ 3.6558 \\ 3.5610 \end{bmatrix}$$

with normal equations

$$\begin{bmatrix} 9 & 0 \\ 0 & 60 \end{bmatrix} \begin{bmatrix} \log a_0 \\ a_1 \end{bmatrix} = \begin{bmatrix} 30.4684 \\ 4.9561 \end{bmatrix}$$

and solution $\log a_0 = 3.3854$ (corresponding to $a_0 = 29.53$) and $a_1 = 0.0826$. One can use many different scalings and models for this problem; the one above is just one if the simplest.

2. Let three points x_1 , x_2 and x_3 be given. Construct formally the Lagrange basis polynomials for interpolation at these three points. **(2p)**

Solution.

$$\begin{aligned} L_1(x) &= \frac{(x - x_2)(x - x_3)}{(x_1 - x_2)(x_1 - x_3)} \\ L_2(x) &= \frac{(x - x_1)(x - x_3)}{(x_2 - x_1)(x_2 - x_3)} \\ L_3(x) &= \frac{(x - x_1)(x - x_2)}{(x_3 - x_1)(x_3 - x_2)} \end{aligned}$$

C. Integration

1. In Table 2 below you find sampled values for a function $f(x)$. Compute

$$\int_0^{0.8} f(x) dx$$

as accurately as possible, and estimate the error.

(4p)

x	0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8
$f(x)$	1.0000	1.1052	1.2214	1.3499	1.4918	1.6487	1.8221	2.0138	2.2255

Table 2: Data for problem **C1**.

Solution. First of all, let us note that for the correctly rounded samples of f above, we have an error at each point of at most $5 \cdot 10^{-5}$. That implies that we can consider f to have an error δf of magnitude at most $5 \cdot 10^{-5}$. This contributes an error to the integral, corresponding to the length of the interval 0.8 times $|\delta f|$, i.e., $4 \cdot 10^{-5}$. Hence we cannot expect to determine the integral to a higher accuracy than this. We try the Trapezoidal rule with extrapolation for $h = 0.4$ and refine the step size from there. We get the following extrapolation scheme:

h	$T(h)$	$\Delta/3$	$S(h)$
0.4	1.241838		
		-0.004072	
0.2	1.229623		1.225552
		-0.001020	
0.1	1.226562		1.225542

Inspecting the correction terms $\Delta/3$, the scheme is perfectly regular. We also see that the truncation error in the final result can be estimated by $10^{-5}/15$, which is far below the best possible accuracy caused by rounded f -values. We therefore accept the value 1.22554 as the best possible, with an error less than or approximately equal to $4 \cdot 10^{-5}$.

D. Nonlinear equations

1. What method does the following MATLAB script use, and what problem does it solve? (2p)

```
x = zeros(2,1);
for i=1:10
    J = [1+sin(x(1))  2; 3+x(2)  x(1)];
    y = [2+x(1)-cos(x(1))+2*x(2) ; 1+3*x(1)+x(1)*x(2)];
    x = x - J\y;
end
x
```

2. What is the value of $\mathbf{x}$ after one iteration? (2p)

Solution. The script solves the nonlinear equation

$$\begin{aligned} 2 + x_1 - \cos x_1 + 2x_2 &= 0 \\ 1 + 3x_1 + x_1x_2 &= 0 \end{aligned}$$

using Newton's method. After one iteration the value of x is $x_1 = x_2 = -1/3$ (starting from zero as the initial guess).

E. Differential equations

1. We want to solve Fisher's equation in the stationary case, i.e., the two-point boundary value problem

$$u'' + u(1 - u) = 0$$

with boundary values $u(0) = 0$ and $u(1) = 1$. Describe in full detail how you introduce a grid (make a drawing, put labels etc.) and discretize the problem using a second order method. Formulate the nonlinear equation system that has to be solved, and be sure to include the boundary conditions. **(3p)**

Solution. Introduce an equidistant grid with N internal points on $[0, 1]$ and two boundary points. The step size is then $\Delta x = 1/(N + 1)$.

The standard second order discretization becomes

$$\frac{u_{j-1} - 2u_j + u_{j+1}}{\Delta x^2} + u_j(1 - u_j) = 0$$

for $j = 2 \dots N - 1$. Accounting for the boundary values, the first equation ($j = 1$) is

$$\frac{-2u_1 + u_2}{\Delta x^2} + u_1(1 - u_1) = 0$$

and the last one ($j = N$) is

$$\frac{u_{N-1} - 2u_N + 1}{\Delta x^2} + u_N(1 - u_N) = 0.$$

These are the N equations in the nonlinear system that we have to solve.

2. The scalar van der Pol equation

$$\ddot{y} + \mu \dot{y}(1 - y^2) + y = 0$$

with initial conditions $y(0) = 0$, $\dot{y}(0) = 1$, and where μ is a parameter, models a nonlinear electrical oscillator. Show how to rewrite this equation as a system of first order differential equations. Remember to clearly specify your variable substitutions and include new initial conditions. **(2p)**

Solution. The standard transformation introduces the variable transformation $z_1 = y$, $z_2 = \dot{y}$. We then get the system

$$\begin{aligned} \dot{z}_1 &= z_2 \\ \dot{z}_2 &= -\mu z_2(1 - z_1^2) - z_1 \end{aligned}$$

with initial values $z_1(0) = 0$ and $z_2(0) = 1$.

3. Let us denote the period of the orbit by T . In order to compute the periodic orbit, one can transform the equation to a two-point boundary value problem on a known interval, say $[0, 1]$, and solve that problem directly. Write down that transformed system, and include boundary conditions that specify that the solution is periodic. Explain your notation and describe in words what a solver has to do, and what information is needed in order to solve the problem. **(3p)**

Solution. We solved this problem in class, and in the second computer assignment the task was the same for a different equation. Because the period T is unknown, we introduce a new independent variable, $x = t/T$, so that the solution over a full period (the interval $[0, T]$) corresponds to the interval $[0, 1]$ (a known period). This means that

$$d/dx = Td/dt,$$

therefore, if we let $'$ denote d/dx , the equation becomes

$$\begin{aligned} z_1' &= Tz_2 \\ z_2' &= -T\mu z_2(1 - z_1^2) - Tz_1 \end{aligned}$$

where the code needs to find z_1 , z_2 and T . Boundary conditions are specified so that the solution is periodic, e.g. $z_1(0) = z_1(1)$ and $z_2(0) = 0$, and additionally we need to specify an initial guess for the solution, e.g. trigonometric function periodic on $[0, 1]$.

F. Quick questions (Only a single line answer or formula is needed in each case.)

1. Let some iterative method be used to solve $f(x) = 0$, and let ε_n be the error in x_n . Define *convergence order*. **(1p)**

Answer. If the error is $\varepsilon_{n+1} \sim \varepsilon_n^p$ as $n \rightarrow \infty$, then the convergence order is p .

2. The Fast Fourier Transform (FFT) is well known for being able to process N data points in only $N \cdot \log_2 N$ operations, if N is chosen correctly. What is this optimal choice? **(1p)**

Answer. It is optimal to choose $N = 2^p$ for some integer p .

3. For the initial value problem $\dot{y} = f(y)$, what is the name of the method

$$y_{n+1} = y_n + \frac{h}{2} (f(y_n) + f(y_{n+1}))$$

and what is its convergence order? **(1p)**

Answer. *Trapezoidal rule*, order $p = 2$.

4. What does the following iteration do and what is the method called?

$$\begin{aligned}\hat{y}_k &:= y_k / \|y_k\|_2 \\ y_{k+1} &:= A\hat{y}_k \\ \sigma_k &:= \hat{y}_k^H y_{k+1}\end{aligned}$$

(1p)

Answer. This is the *power method* for solving the eigenvalue problem $Ay = \lambda y$.

5. Let f be a function that can be evaluated at any point x . Give a second order difference formula for approximating the first derivative $f'(x)$. (1p)

Answer.

$$f'(x) \approx \frac{f(x+h) - f(x-h)}{2h}$$

LYCKA TILL!